The Concept of Causality in Physics

In the fight currently raging about the meaning and validity of the Law of Causality in modern physics, every attempt to clarify the conflicting opinions must begin with the statement that in this connection everything depends on a clear understanding of the sense in which the word “causality” is used in the science of physics. To be sure, it is agreed *a priori* that whenever a reference is made to a “causal relationship” between two successive events or occurrences, this term is understood to designate a certain regular connection between them, calling the earlier one the *cause*, and the latter one the *effect*. But the question is: What constitutes this specific type of connection? Is there any infallible sign to indicate that a happening in nature is causally determined by another?

It follows from the numerous inquiries heretofore undertaken into this question that the best and safest way to approach a clear answer is to relate the question to the possibility of making accurate predictions. In fact, there can be no more incontestable way to prove the causal relationship between any two events than to demonstrate that from the occurrence of one it is always possible to infer in advance the occurrence of the other. This point was quite familiar to the farmer who gave a visual demonstration to some sceptical peasants of the causal relationship between artificial fertilizers and the fertility of the soil, by intensively fertilizating his clover fields in certain narrow strips having the form of letters, so as to make the following sentence appear: “This strip was fertilized with calcium sulfate.”

Therefore, I want to base all our subsequent considerations on the following simple proposition, equally applicable outside of the realm of physics: *“An occurrence is causally determined if it can be predicted with certainty.“* Of course, this sentence is meant to express only that the possibility of making an accurate prediction for the future constitutes an infallible criterion of the presence of a causal relationship, but not that it is synonymous with the latter. I need to mention merely the well known example that we can predict with a certainty while it is still day the coming of night, and yet this does not make day the cause of night.

But conversely, it also often happens that we assume the presence of a causal relationship even in cases where there is no question at all of the possibility of making accurate predictions. Just think of weather forecasts. The unreliability of weather prophets has become proverbial, and yet there is hardly any trained meteorologist who does not consider the atmospheric processes to be causally determined. All these considerations indicate that in